Year 13 Further Maths - Collisions

Similar questions - Oblique impact, impulsive tension

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 Two smooth circular discs A and B of masses 4 kg and 2 kg move on a smooth horizontal plane and collide. Just before the collision, A has speed 6 $\mathrm{m\,s^{-1}}$ at 30° to the line of centres and B has speed 2 $\mathrm{m\,s^{-1}}$ along the line of centres towards A. The coefficient of restitution is 0.5.

(a) Find the speed and direction of A after the collision. **[7]**

(b) Find the total loss of kinetic energy. **[4]**

2 Two smooth circular discs A and B of masses 2 kg and m kg move on a smooth horizontal plane. Just before they collide, A has speed 5 $\mathrm{m\,s^{-1}}$ at an angle α to the line of centres where $\tan\alpha = \frac{4}{3}$, and B has speed 4 $\mathrm{m\,s^{-1}}$ at 60° to the line of centres, on opposite sides of it so that their components along the line of centres are towards each other. After the collision B moves perpendicular to the line of centres.

(a) Show that $3 \leqslant m \leqslant 8$. **[6]**

(b) Given that $m = 5.5$, find the total loss of kinetic energy. **[5]**

3 Two particles A and B, of masses 2 kg and 1 kg, are joined by a light inextensible string of length 1 m and rest on a smooth horizontal table at a distance 0.6 m apart. B is projected at 5 $\mathrm{m\,s^{-1}}$ perpendicular to AB.

(a) Show that the magnitude of the impulse in the string at the instant it becomes taut is $\frac{8}{3}$ N s. **[5]**

(b) Find the kinetic energy of B just after the string becomes taut. **[3]**

(c) If A were much heavier than B, describe the approximate motion of B and give its kinetic energy in that case. **[2]**

4 Two smooth discs A (mass 5 kg) and B (mass 10 kg) collide. Before the collision A has speed 10 $\mathrm{m\,s^{-1}}$ at an angle α to the line of centres where $\cos\alpha = 0.8$, and B has speed 5 $\mathrm{m\,s^{-1}}$ along the line of centres towards A. After the collision the velocity of A is perpendicular to its velocity before the collision.

(a) Find the coefficient of restitution between A and B. **[7]**

(b) Find the total loss of kinetic energy. **[4]**

Question 1

(a) Along the line (towards B positive): $u_A = 6\cos 30^\circ = 5.196$, $u_B = -2$; perpendicular to the line A has 3, unchanged. Momentum: $4v_A + 2v_B = 4(5.196) - 4 = 16.78$. Restitution: $v_B - v_A = 0.5(7.196) = 3.598$. So $6v_A = 16.78 - 7.196$, $v_A = 1.598$, $v_B = 5.196$. Speed of $A = \sqrt{1.598^2 + 3^2} = 3.40\,\text{m s}^{-1}$ at $\tan^{-1}\frac{3}{1.598} = 62.0^\circ$ to the line of centres (continuing towards B's side)

(b) KE before $= \frac{1}{2}(4)(36) + \frac{1}{2}(2)(4) = 76$ J; after $= \frac{1}{2}(4)\left(1.598^2 + 9\right) + \frac{1}{2}(2)(5.196)^2 = 23.1 + 27.0 = 50.1$ J. Loss $= 25.9$ J

Question 2

(a) Along the line (A positive): $u_A = 3$, $u_B = -2$ ($4\cos 60^\circ$); B's final component along the line is 0, so $2(3) + m(-2) = 2v_A$ and $0 - v_A = e(3+2)$, $v_A = -5e$. So $6 - 2m = -10e$, $m = 3 + 5e$. Since $0 \leqslant e \leqslant 1$: $3 \leqslant m \leqslant 8$

(b) $e = 0.5$, $v_A = -2.5$; B keeps its perpendicular component $4\sin 60^\circ = 3.46$. KE before $= \frac{1}{2}(2)(25) + \frac{1}{2}(5.5)(16) = 69$ J. After $= \frac{1}{2}(2)\left(2.5^2 + 4^2\right) + \frac{1}{2}(5.5)(3.46^2) = 22.25 + 33.0 = 55.25$ J. Loss $= 13.8$ J

Question 3

(a) When taut, B is 1 m from A and has moved 0.8 m, so the string makes angle ϕ with AB where $\cos\phi = 0.6$. B's velocity $(0,5)$ has component $5\sin\phi = 4$ along the string and $5\cos\phi = 3$ perpendicular to it. Along the string: $1(4) = (2+1)w$ so $w = \frac{4}{3}$; impulse $= 2w = \frac{8}{3}$ N s

(b) B has $\frac{4}{3}$ along the string and 3 perpendicular: KE $= \frac{1}{2}\left(\frac{16}{9} + 9\right) = 5.39$ J

(c) A effectively stays at rest, the string removes the $4\,\text{m s}^{-1}$ component, B moves in a circle of radius 1 m about A at speed $3\,\text{m s}^{-1}$ with KE 4.5 J

Question 4

(a) Along the line A: 8, perpendicular 6; B: -5. After: $A = (c,6)$ with $(c,6)\cdot(8,6) = 0$ so $c = -4.5$. Momentum: $5(8) + 10(-5) = 5(-4.5) + 10v_B$, $v_B = 1.25$. $e = \dfrac{1.25 - (-4.5)}{8 - (-5)} = \dfrac{5.75}{13} = 0.442$

(b) KE before $= \frac{1}{2}(5)(100) + \frac{1}{2}(10)(25) = 375$ J; after $= \frac{1}{2}(5)(20.25 + 36) + \frac{1}{2}(10)(1.5625) = 140.6 + 7.8 = 148.4$ J; loss $= 227$ J